

Electrophoresis investigation - Answers

DNA profiling of Humpback Whales

(30 marks)

DNA profiling is a process that uses DNA to show relatedness or identity of individual humans, plants, or animals. It is used in forensics, anthropology, and conservation biology not only to determine the identity of individuals but also to determine relatedness. This process has been used to free innocent suspects, reunite children with their relatives, identify stolen animals, and prove that whale meat has been substituted for fish in sushi. It is used in times of war to help identify the remains of soldiers killed in combat. It is also being used to find genetic linkages to inherited diseases. In addition, scientists are learning a great deal about our evolutionary history from DNA analysis.

The Scenario:

For three years a team of researchers have been watching three different Humpback whale pods off the coast of Exmouth in Western Australia. They are especially interested in the genetic diversity of these whales. As the numbers of humpback whales are decreasing, determining the genetic diversity of these populations is increasingly more important. In particular, this team is interested in identifying the parents of one of the new baby whales, Kara. Through their observations of nursing behavior they have matched Kara with her mother; however, none of their observations have helped them link Kara to her father. Today you are going to help them figure this out!



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Humpback Whale Pod Information:

Pod	Gender	DNA sample
1	Female (mother)	1
1	Female (Kara)	2
1	Male A	3
2	Male B	4

Materials

DNA from Mother (1), Kara (2), and the two male whales (3 and 4)
Gel electrophoresis apparatus and power supply

Procedure:

Follow the instructions provided by you teacher for setting up and using the gel electrophoresis equipment.

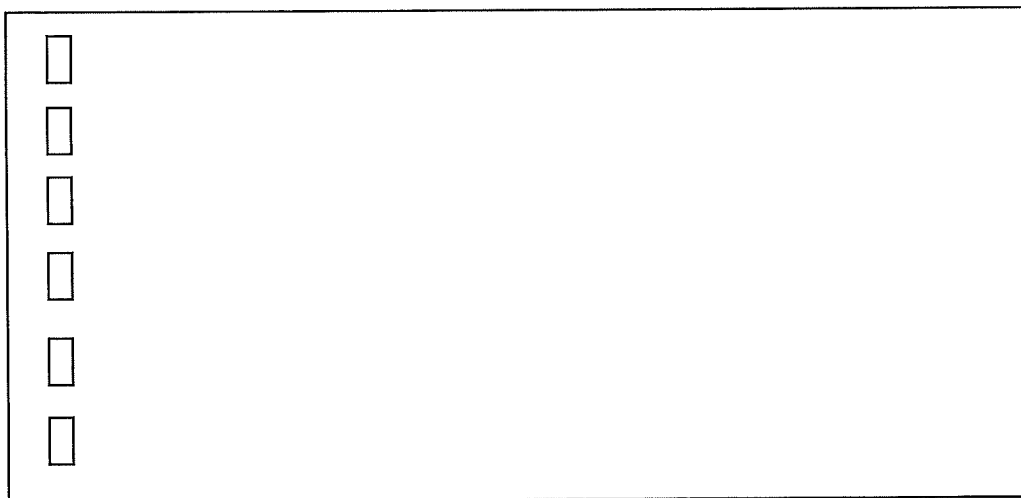
1. Each student must effectively add 1 whale DNA sample each into the correct sample wells according to the paper template provided by the teacher. These samples will be 'run' in the electrophoresis tank for you to analyse later.
2. Pipet two more samples (as per the paper template instructions provided by your teacher) into the gels contained in the take-away contains.

With all samples, will be assessed on your pipetting technique and accuracy. (3 marks)

Analyzing the DNA data

On the diagram below label which wells contain which DNA samples. (1 mark)

After the DNA samples have sufficiently separated, use either tank's results (or the supplied photo) to draw a diagram of the profiles of each sample in the test. Be as accurate as possible in drawing the bands. A ruler may be useful for this. (2 marks)



Does the electrophoresis results confirm which whales are the parents of Kara? Outline the evidence clearly in your answer. (4 marks)

___ The position of two of the bands in Kara's profile clearly match up/line up with the band found in the Mother's profile (1). Kara's second band also lines up with a band from Male B (1). None of Kara's bands line up with Male A (1). This would indicate that Male B is the father of Kara (1). _____

Answer the following questions

1. DNA sample from a whale

Suggest a viable method that could be used to obtain DNA samples from a whale that is swimming in the ocean. (Remember that it is usually not possible to get close enough to take DNA directly). (1 mark)

Collect a faeces sample that has been released by the whale and separate the epithelial
cells OR Use a long pole with a container attached to the end and from a boat,
get close enough to obtain material that the whale releases through its blowhole, which should
contain epithelial cells.

OR any other method deemed viable at the markers discretion

2. DNA samples in general

- a) Name a likely type of cell that DNA can be obtained from? (1 mark)

skin cell, blood cell, cheek cell, sperm cell, hair follicle, et.al.

- b) Which areas of the DNA strand would be prepared for the DNA profiling and what would these areas contain? (2 marks)

Non-coding regions that contain specific short tandem repeats (STRs)

- c) How are these samples 'separated' from the rest of a cell's DNA? (1 mark)

restriction enzymes cut at specific recognition sites

- d) Briefly explain how these DNA samples taken from two individuals create different banding profiles observed in gel electrophoresis. (2 marks)

STRs are made up of a string of a particular nucleotide base sequence that has
been repeated different amounts of times in different individuals (1). These different
lengths of STRs will migrate through gel at different rates resulting in a unique banding
pattern that is different for each individual tested (1).

3. If there is only a very small sample of DNA available, it can be amplified to produce enough DNA for DNA profiling;
- Name this process: **polymerase chain reaction (1/2)** ____ (PCR not accepted) ____ (1 mark)

 - Components/ingredients required in this process (4): (2 marks)
DNA template (1/2), primers (1/2), free nucleotide bases (1/2), Taq polymerase (1/2) ____
 - Name the steps involved in this process (3) and a brief explanation of each step. (3 marks)
denaturing (1/2): template DNA heated to separate into two single strands (1/2)
annealing (1/2): cooled to allow primers to attach to template (1/2)
extending (1/2): heated to allow new strand to be made (1/2)
4. Once DNA samples have been added to the wells of an electrophoresis tank it is then switched on.
- Outline what happens next to the samples and why? (3 marks)
Electric current/circuit causes -ve DNA to migrate to the +ve end of the gel (1)
Smaller fragments move faster/further than large fragments (1)
Electric circuit is turned off and fragments stop moving forming bands within a profile (1)
 - As DNA is invisible, describe one technique used to make the DNA profile visible. (2 marks)
radioactive marker (1) and x-ray film (autoradiograph) (1)
fluorescent marker (1) and UV light (1)
DNA stain used (1) before or after electrophoresis (1)
DNA stain added to gel (1) to attach to the DNA as it moves (1)
(1 of for a max. of 2 marks)
5. Poppy the pigeon is a racing pigeon and her chick has been stolen. The owner believes that a rival has stolen her chick. The rival claims it is a chick from one of his own pigeons called Patsy. Use the table to determine who the chick belongs to and explain why. (2 marks)

Genetic fingerprints for the three pigeons

satellite	Number of satellite repeats at each locus		
	chick	poppy	patsy
A	6, 12	5, 12	10, 6
B	7, 4	4	8, 5
C	9, 5	9, 13	6, 11

Identify/explain how they know (1)

Relate to inheritance of STRs 1 from Mum and 1 from Dad (1)